ELIO VINCIGUERRA

Catania, Italy
elio.vinciguerra@phd.unict.it

EDUCATION

Ph.D. in Systems Engineering, Energy, Computer Science, and Telecommunications — *Aug. 2023 - Jul. 2026*
University of Catania, Italy

Master's Degree in Computer Engineering — *Oct. 2021 - Jul. 2023*
University of Catania, Italy
Thesis: *Analysis and Optimization of Indoor Localization Algorithms for HPC Architectures*
Final grade: 108/110

Bachelor's Degree in Computer Engineering — *Oct. 2018 - Nov. 2021*
University of Catania, Italy
Thesis: *Evaluation of the Energy-Latency Trade-off in Deep Neural Network Accelerators*
Final grade: 95/110

Diploma in Computer Science and Telecommunications — *Sept. 2013 - Jul. 2018*
I.I.S I.T. G. Marconi, Italy
Final grade: 94/100

EXPERIENCE

KTH Royal Institute of Technology — Apr. 2026 - May 2026
Visiting PhD Student — *Stockholm, Sweden*

- Use of the NoCDAS framework and integration of Transformers models and computational NoC paradigm within it.

KTH Royal Institute of Technology — Sept. 2025 - Jan. 2026
Visiting PhD Student — *Stockholm, Sweden*

- Use of the CGRA SiLago framework and integration of Deep Reinforcement Learning algorithms within it.

University of Catania — Nov. 2023 - Jul. 2026

- Computer Architecture Course Teaching Assistance.
- Fundamentals of Computer Science Course Teaching Assistance.
- IoT Systems and Technologies Course Teaching Assistance

Flazio S.r.l. — Nov. 2017 - Dec. 2017
Intern — *Catania, Italy*

- Web Design, website development using the Flazio platform.
- Social media integration.

BaxEnergy S.r.l — Jul. 2017 - Aug. 2017
Intern — *Catania, Italy*

- Quality control of company software programs.

Cultural Exchange Program at Polytechnic Institute Jin Yu Sept. 2016

LANGUAGE PROFICIENCY

Italian	Native
English	Intermediate (B2)
Spanish	Intermediate (B1)
Chinese	Basic (A1)

TECHNICAL STRENGTHS

Programming Languages	Python, C, C++, Java, Go, C#
Web Technologies	HTML, CSS, PHP
Databases	SQL (MySQL, PostgreSQL, Microsoft SQL)
Tools	Git, VS Code, IntelliJ, Eclipse

PUBLICATIONS

1. E. Vinciguerra, E. Russo, G. Ascia, and M. Palesi, “CHAOS: Controlled hardware fault injector system for gem5,” 2026. arXiv: 2602.02119 [cs.AR]. [Online]. Available: https://arxiv.org/abs/2602.02119.

2. E. Vinciguerra, M. Palesi, and G. Ascia, “Improving Machine Learning Anomaly Detection for Safety-Critical RISC-V Automotive Systems,” in Proceedings of the CPS Summer School PhD Workshop 2025, C. Pilato et al., Eds., vol. 4106. Aachen, Germany: CEUR-WS.org, 2025. [Online]. Available: https://ceur-ws.org/Vol-4106/short1.pdf.

3. E. Russo, E. Vinciguerra, M. Palesi, D. Patti, G. Ascia and V. Catania, “TeleSABRE: Heuristic Layout Synthesis in Multi-Core Quantum Systems with Teleport Interconnect,” 2025 IEEE International Conference on Quantum Computing and Engineering (QCE), Albuquerque, NM, USA, 2025, pp. 749-758, doi: 10.1109/QCE65121.2025.00086.

4. E. Vinciguerra, E. Russo, M. Palesi, G. Ascia, “An Anomaly Detection Model for RISC-V in Automotive Applications: A Domain-Specific Accelerator Perspective,” 2025 33rd Euromicro International Conference on Parallel, Distributed and Network-Based Processing (PDP), Turin, Italy, 2025, pp. 317-320, doi: 10.1109/PDP66500.2025.00051.

5. E. Vinciguerra, E. Russo, M. Palesi and G. Ascia, “Data-Driven Simulation Based Fault Detection in Automotive RISC-V Applications,” *2024 IEEE 17th International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoC), Kuala Lumpur, Malaysia, 2024*, pp. 509-516, doi: 10.1109/MCSoC64144.2024.00089.

6. E. Russo, E. Vinciguerra, G. Ascia and D. Panno,“DNN Hardware Accelerator Selection for Feasible Deployment of MARL-Based MAC Protocols in Industrial IoT Networks,” 2024 IEEE 8th Forum on Research and Technologies for Society and Industry Innovation (RTSI), Milano, Italy, 2024, pp. 443-448, doi: 10.1109/RTSI61910.2024.10761172.

7. E. Vinciguerra, E. Russo, M. Palesi, G. Ascia and H. Rafique, “Improving LSTM-based Indoor Positioning via Simulation-Augmented Geomagnetic Field Dataset,” *2024 IEEE International Conference on Mobility, Operations, Services and Technologies (MOST)*, Dallas, TX, USA, 2024, pp. 251-259, doi: 10.1109/MOST60774.2024.00034.